Задания, которые необходимо выполнить для получения зачета.

1. Выполнить задания по предложенной статье.
2. Выбрать 3 **статьи** по специальности (каждая – не менее **10.000** знаков без пробелов).
   По каждой статье:
   1. Выполнить перевод на русский язык.
   2. Выписать не менее 30 слов и выражений, отметить их в тексте статьи.
   3. Задание 1 (TASK 1)
   4. Аннотация
   5. Устная презентация содержания статьи(5-7 минут речи).

TASK 1

The overall structure of a research paper

Title

Author(s)

Abstract

Key words

- Body (see below)
- References
- About the authors
- Acknowledgement (optional)

Body The standard IMRD (or IMRaD) format (see the scheme below):

- **Introduction**
  - to provide the rationale for the paper moving from general discussion of the topic to the particular question of hypothesis being investigated
  - - to attract interest in the topic – and hence readers
- **Methods (and materials)**
  - to describe, in various degrees of detail, methodology, materials (or subjects), and procedures. This is the narrowest part of the research paper
- **Results**
  - - to describe the findings, give commentary
- **Discussion**
  - to offer a generalised account of what has been learned in the study. This is usually done through a series of "points" some of which refer back to the statements made in the Introduction

Step 1: Navigating and positioning a text using publication details

Task 1: Highlight or underline the following information:

- Name of the journal
- The volume number of the journal
- The issue number of the volume (if found)
- The year of publication
- The first and the last page numbers of the article
- The title of the article
- The names (surnames and initials) of the authors.

Step 2: Genre, audience, and purpose

- Task 2: Think about the following questions:
  - 1. What type of article* is this?
  - a research article
  - a review article
  - another type of article (e.g. a theoretical paper)
- 2. Who do you think it is written for (more than one answer is reasonable)?
  - graduate students or undergraduate students
  - researchers and/or lecturers and/or professors
  - novices or experts in carbon sequestration
  - a national audience or an international audience
- Look at the title of the article.
  - Why has the article been written?
  - What is the object of the article?
  - What does the article review?

*Note: Science Direct has 3 categories of article: 'research' articles; 'review' articles; other articles. In a list of search results you will see the words 'research' or 'review' under the title of

the article. If you do not see a word under the title this means that the article is categorised as ‘other’.

STEP 3

Study the whole text and answer the questions.

a. What is the background and aim of the investigation?
b. What was done in the investigation?
c. What was found/discovered?
d. What do the Results mean?
e. What literature sources are referenced in the paper?

TASK 2

АННОТАЦИЯ -

это краткое содержание статьи с элементами анализа.

Структура аннотации:

1. **Вступление.**

The article I'm going to give a review of is taken from… — Статья, которую я сейчас хочу проанализировать из…

The headline of the article is — Заголовок статьи…

The author of the article is… — Автор статьи…

It is written by — Она написана …

The article under discussion is … — Статья, которую мне сейчас хочется обсудить, …

2. **Тема. Логические части.**

The topic of the article is… — Тема статьи

The key issue of the article is… — Ключевым вопросом в статье является

The article under discussion is devoted to the problem… — Статью, которую мы обсуждаем, посвящена проблеме…

The author in the article touches upon the problem of… — В статье автор затрагивает проблему….

The article under discussion may be divided into several logically connected parts which are… — Статья может быть разделена на несколько логически взаимосвязанных частей, таких как…

3. Краткое содержание.

The author starts by telling the reader that — Автор начинает, рассказывая читателю, что

At the beginning of the article the author — В начале истории автор

describes — описывает

The article begins (opens) with a (the)

description of **—** описанием

statement — заявлением

introduction of — представлением

<u>Then / After that</u> the author

gives examples – даёт примеры

depicts — изображает

touches upon — затрагивает

explains — объясняет

introduces — знакомит

mentions — упоминает

recalls –вспоминает

makes a few critical remarks on — делает несколько критических замечаний о

dwells on — останавливается на

points out — указывает на то

generalizes — обобщает

reveals — показывает

exposes — показывает

accuses/blames -обвиняет

gives a summary of -дает обзор

In conclusion the author says / makes it clear that…/ gives a warning that… - В заключение автор говорит / проясняет, что … / дает предупреждение, что …

At the end of the article the author sums it all up by saying … - В конце рассказа автор подводит итог всего этого, говоря …

The author concludes by saying that../ draws a conclusion that / comes to the conclusion that - В заключение автор говорит, что….. / делает вывод, что…. / приходит к выводу, что…..

TASK 3

УСТНОЕ СООБЩЕНИЕ (ПРЕЗЕНТАЦИЯ)

1.Opening

People tend to remember openers more than any other part of a presentation, except perhaps for the closing remarks.
Andrew Leigh, Maynard Leigh Associates, www. maynardleigh. co.uк

The secret *is:* have a good beginning, a good ending and keep them as close together as possible!
Ceorge Burns, actor and comedian

Opening: According to communication expert Andrew Leigh, there are four stages to opening a presentation.

1 *Capture your audience's interest instantly*

2 *Explain what they will gain from the talk*

3 *Show them you have the authority to speak*

4 *Give them a route map of the presentation*

USEFUL PHRASES

1. Начните презентацию с приветствия и благодарности за то, что все пришли:

- *Good morning, ladies and gentlemen*
- *Good afternoon, ladies and gentlemen*
- *Good afternoon, everybody*
- *Hello, everyone*
- *Welcome everyone*
- *I'd like to start by thanking you all for coming*
- *It's great to see so many fresh faces here today*
- *Let me begin by welcoming you all of you here today*
- *It's great to be here with you all*
- *Thank you for coming out today*

2. Переходим к рассказу о себе и о цели своего доклада:

- *I'm John Smith*
- *My name is John Smith*

- *I am going to talk today about...*
- *The purpose of my presentation is to introduce our new range of..*
- *There are three main areas I want to look at today*

3. Опишите слушателям, какая программа ждет их во время вашего выступления:

- *To start with I'll describe*
- *Then I'll mention some of the problems we've encountered and how we overcame them.*
- *After that I'll consider the possibilities for further growth next year.*
- *Finally, I'll summarize my presentation.*
- *Firstly ... secondly ... thirdly ... lastly ...*
- *First of all ... then ... next ... after that ... finally ...*
- *To start with ... later ... to finish up ...*

2. Body: Smooth Structure

4. Здесь мы рассмотрим, как начать основную часть презентации на английском:

- *I'd like to start by*
- *Let's begin by ...*
- *First of all, I'll ...*
- *Starting with ...*
- *I'll begin by .*

5.Если вы закончили очередную логическую часть, стоит это обозначить:

- *Well, I've told you about ...*
- *That's all I have to say about ...*
- *We've looked at ...*
  So much for ...

6. Начиная новую часть презентации, предупредите об этом слушателей, чтобы они не теряли нить выступления:

- *Now we'll move on to*
- *Let me turn now to*
- *Next*
- *Turning to*
- *I'd like now to discuss*
- *Let's look now at ... -*

7. После того, как вы рассказали основную информацию, необходимо её проанализировать:

- *Where does that lead us? –*
- *Let's consider this in more detail ...*
- *What does this mean for ...*
- *Translated into real terms ...*

Чтобы информация лучше воспринималась, приводите больше примеров:

- *For example,*
- *A good example of this is ...*
- *As an illustration, ... –*
- *To give you an example, ...*
- *To illustrate this point ...*

-

8. Если вам задали вопрос во время презентации, сориентируйте аудиторию, когда вы на него сможете ответить:

- *We'll be examining this point in more detail later on*
- *I'd like to deal with this question later, if I may .*
- *.. I'll come back to this question later in my talk –*
- *I won't comment on this now ...*

3. Closing:

According to communication expert Sonya Hamlin, Harward Law School presentations coach, there are four techniques to close a presentation

1. *Summary*
   - Let's take a look back at what we've spoken about this morning b I'm reminded of the words of..
   - . Here, at a glance, are the main points I've made ...
   - So, how to sum up?
   - In a nutshell, then
2. *Wisdom*
   - I'm reminded of the words of …..
   - We have a saying where I come from ...
3. *Action*
   - Now let's get out there and ...!
   - I'm counting on you to ...
4. *Emotion*
   - If you take just one thing from this talk, take this ...

- In the end, this is what matters

9. Заканчиваем презентацию английскими фразами, обозначающими подведение итогов:
- *In conclusion, ...*
- *Right, let's sum up, shall we?*
- *Let's summarise briefly what we've looked at ...*

10. *Finally, let me remind you of some of the issues we've covered ...*

Обязательно поблагодарите публику за внимание в конце выступления:
- *Many thanks for your attention.*
- *May I thank you all for being such an attentive audience.*

Перед тем, как завершить презентацию, объявите слушателям, что вы готовы ответить на их вопросы:
- *Now I'll try to answer any questions you may have.*
- *Are there any questions?*
- *Do you have any questions?* *Are there any final questions?*

Chen, L., Patel, R., Volkov, S., & Nakamura, K. (2025). Latency reduction in IoT networks through edge AI deployment: A comparative study. Journal of Edge Computing and IoT Systems, 12(3), 145–152.

LATENCY REDUCTION IN IOT NETWORKS THROUGH EDGE AI DEPLOYMENT: A COMPARATIVE STUDY

Chen, L.; Patel, R.; Volkov, S.; Nakamura, K.

Abstract (short version)

This study compares cloud-only, edge-only, and hybrid AI inference approaches to reduce latency in IoT networks. Experiments on 50 ESP32-S3 nodes with a quantized MobileNetV3 model showed that edge-only processing cut mean latency from 187.6 ms to 23.4 ms while maintaining accuracy (~94 %). The hybrid approach offered a balanced trade-off with 61.2 ms latency and the lowest energy consumption (12.3 mJ per inference). Results confirm edge AI's effectiveness for low-latency IoT applications.

Keywords: edge AI; IoT networks; latency reduction; edge computing; hybrid architecture; energy efficiency.

Introduction

The rapid proliferation of Internet of Things (IoT) devices has placed unprecedented strain on centralized cloud infrastructure. By 2026, an estimated 30 billion connected devices generate continuous streams of sensor data, much of which requires real-time processing for applications such as autonomous vehicles, industrial monitoring, and smart healthcare. Traditional cloud-centric architectures introduce network latency that can exceed 200 ms, which is unacceptable for time-critical workloads. Edge computing, which brings computation closer to the data source, offers a promising alternative. This study investigates whether deploying lightweight AI inference models directly on edge nodes significantly reduces end-to-end latency compared to cloud-based processing, without sacrificing inference accuracy.

Methods

We conducted a controlled experiment using a testbed of 50 IoT nodes equipped with ESP32-S3 microcontrollers and a local edge server based on an NVIDIA Jetson Orin Nano. Three inference configurations were evaluated: (1) full cloud processing via AWS EC2 (g5.xlarge), (2) edge-only inference using TensorFlow Lite Micro on the Jetson, and (3) a hybrid approach where initial filtering occurred on the edge node and complex inference was offloaded to the cloud. The workload consisted of real-time image classification tasks using a quantized MobileNetV3 model. Each configuration was tested over 1,000 inference cycles. We measured end-to-end latency (ms), inference accuracy (%), and energy consumption per inference (mJ). Statistical significance was assessed using paired t-tests with a significance level of $p < 0.05$.

Results

The edge-only configuration achieved a mean latency of 23.4 ms (SD = 4.1), compared to 187.6 ms (SD = 22.3) for cloud-only and 61.2 ms (SD = 8.7) for the hybrid approach. The difference between edge-only and cloud-only was statistically significant ($p < 0.001$). Inference accuracy remained comparable across all configurations: 94.2% (cloud), 93.8% (edge), and 94.0% (hybrid), with no statistically significant difference ($p = 0.34$). Energy consumption per inference was lowest for the hybrid approach at 12.3 mJ, while edge-only consumed 18.7 mJ and cloud-only 31.2 mJ (the latter due to radio transmission overhead).

Discussion

The results demonstrate that edge AI deployment reduces latency by approximately 87% compared to cloud-only processing while maintaining equivalent inference accuracy. This finding aligns with recent studies showing that quantized models can retain accuracy within 1% of their full-precision counterparts. The hybrid configuration offers a compelling middle ground: latency is reduced by 67% relative to cloud-only, and energy consumption is the lowest of the three, suggesting that intelligent task partitioning can optimize both speed and power efficiency. However, edge nodes face hardware constraints — memory and thermal limits may restrict model complexity. Future work should explore dynamic model partitioning, where the inference pipeline adapts in real time based on current network conditions and node load. Additionally, security implications of

distributing AI models across edge devices warrant further investigation, as each node becomes a potential attack surface.

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